# Lecture 32 — Determinants, geometry, and invertibility

**Fri Nov 6**

Last updated: October 8, 2026.

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Determinants (class notes). Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

## Learning objectives

By the end of class, you should be able to:

1. Compute determinants of $2 \times 2$ and simple triangular matrices.
2. Interpret determinant magnitude as area or volume scaling.
3. Connect determinant zero with lost dimension and noninvertibility.
4. Recognize the determinant of a triangular matrix from its diagonal.
5. Connect determinants to the coming characteristic equation.

## Warm-up

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

(a) What does $x \mapsto Ax$ do to the unit square? What is the area of the image?

(b) Are the columns of $B$ independent? Guess whether $B$ is invertible.

## Fill in

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$\det A = ad - bc.$$

Geometrically, $|\det A|$ is the _____ scaling factor of the unit square under $x \mapsto Ax$ (signed area encodes orientation).

**Triangular / diagonal.** If $T$ is upper or lower triangular (including diagonal),

$$\det T = _____.$$

**Invertibility.** For square $A$:

$$A \text{ invertible} \quad \iff \quad \det A \neq ____.$$

If $\det A = 0$, the image of the unit cube/square is _____ (lost dimension): columns are dependent, $Ax = b$ fails for some $b$, and $N(A)$ is nontrivial.

**Product rule (preview).** $\det(AB) = (\det A)(\det B)$. Useful later for compositions.

**Coming attraction.** Eigenvalues solve $\det(A - \lambda I) = 0$ — the **characteristic equation**. Today's det is the tool; Week 13 builds the theory.

## Worked in class

**W1.** Compute.

(a) $\det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

(b) $\det \begin{bmatrix} 1 & 4 & 0 \\ 0 & -2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$

(c) $\det \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$

**W2.** Area scaling. Let $A = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$. Compute $\det A$. Where do $e_1$ and $e_2$ go? What area does the unit square map to?

**W3.** Zero determinant. Explain, in one sentence each, why $\det \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = 0$ means (i) noninvertible and (ii) the image of the unit square has area zero.

**W4.** Link to Monday. For invertible $2 \times 2$ $A$, the formula for $A^{-1}$ has $\det A$ in the denominator. What goes wrong if $\det A = 0$?

**W5.** Characteristic preview. For $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, expand $\det(A - \lambda I)$ as a quadratic in $\lambda$. (No need to "solve eigenvalues" yet — just form the polynomial.)

## Your turn

**Y1.** Compute $\det \begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix}$ and $\det \begin{bmatrix} -1 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 1 & 2 \end{bmatrix}$.

**Y2.** Let $A = \mathrm{diag}(2, -1)$. What is $|\det A|$? Interpret as area scaling of the unit square.

**Y3.** True or false: if $\det A = 0$ for square $A$, then $Ax = b$ has either no solution or infinitely many solutions for every $b$.

## Answers (Your turn)

Y1. $\det \begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix} = 12 - (-2) = 14$. Lower triangular: $\det = (-1)(5)(2) = -10$.

Y2. $|\det A| = |(2)(-1)| = 2$: the unit square is mapped to a rectangle of area 2 (stretched by 2 horizontally and reflected/flipped vertically).

Y3. **True.** Singular means no unique solution for any $b$: if $b \in C(A)$ there are infinitely many solutions; otherwise none.